

Prathamesh Ganesh Jadhav

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AI/ML ENGINEER · DATA SCIENTIST · GENAI ENGINEER

Engineering-focused **AI/ML developer** with hands-on experience building **production-grade vector databases, RAG pipelines, and LLM-integrated APIs** from first principles. Skilled in scalable low-latency infrastructure — **custom HNSW indexes, hybrid semantic + BM25 retrieval** — and systems-level engineering including **multi-threaded network analysis**. Delivered quantified impact in an internship; 80+ open-source commits across production projects. Targeting **GenAI, data science, and applied AI** roles.

TECHNICAL SKILLS

Languages Python, JavaScript, TypeScript, Java, C	AI / ML RAG Pipelines, Vector Search, HNSW, BM25, TensorFlow, Scikit-learn, HuggingFace, Transformers, LLM Integration	Data Analytics Pandas, NumPy, Matplotlib, Plotly, Preprocessing
Backend / APIs Flask, FastAPI, REST APIs, SSE Streaming, Multithreading	Databases SQLite, MongoDB, MySQL, PostgreSQL, Redis	Tools & Cloud Git, Docker, AWS, VS Code
Core CS DSA, DBMS, OOP, Operating Systems, Computer Networks		

EXPERIENCE

Data Science Intern — Uptricks Services Pvt. Ltd. Oct 2024 – Dec 2024 · Remote

- **Designed and automated** end-to-end data preprocessing pipelines in Python, reducing manual data-cleaning overhead by ~35% and standardising ingestion workflows across 4+ client datasets.
- **Built ML workflows** using Scikit-learn — cross-validation tuning and feature importance analysis improved model accuracy by 12%; Matplotlib/Plotly dashboards delivered actionable insights across 2 live client projects.
- **Integrated REST APIs** to automate data ingestion, cutting reporting cycles by ~40%; collaborated in an agile team of 6 to deliver data-driven recommendations on schedule.

FEATURED PROJECTS

NuroSearch — Custom Vector Database with Full-Stack RAG Pipeline Feb 2026 - Mar 2026 ■

<https://github.com/Prathamesh-Jadhav04/NuroSearch>

Python · Flask · SQLite · Redis · Ollama · HNSW · KD-Tree · BM25 · Plotly.js · REST APIs · SSE

- **Engineered** a production-grade vector database from scratch implementing HNSW and KD-Tree indexing in pure Python, achieving sub-10 ms nearest-neighbour retrieval on a 50 K-document corpus.
- **Architected** a hybrid search engine combining dense semantic retrieval with BM25 sparse keyword scoring, improving retrieval relevance by ~28% over pure vector search across benchmark queries.
- **Built** a complete end-to-end RAG pipeline — chunking, embedding, indexing, retrieval, and grounded LLM generation via Ollama — processing 1 K+ queries/day as a unified API-driven system.
- **Designed** Flask REST API with SSE streaming for real-time LLM token delivery supporting 20 concurrent clients; Redis caching reduced retrieval latency by ~60%; 40+ open-source GitHub commits with PR-based reviews.

DPI Engine — Multi-Threaded Deep Packet Inspection System

Apr 2026 - May 2026 ■

<https://github.com/Prathamesh-Jadhav04/DPI-Engine>

Python · Multithreading · Packet Analysis · PCAP · TLS/SNI · Network Protocols

- **Developed** a multi-threaded DPI engine achieving 500+ packets/sec concurrent live/offline capture using producer-consumer architecture; validated against 5 public network-capture datasets.
- **Implemented** TLS SNI extraction from ClientHello handshakes enabling hostname-based application identification without payload decryption, achieving 95%+ classification accuracy on encrypted streams.
- **Designed** a rule-based filtering engine with configurable signature matching — IP/port/protocol blocklisting and pattern-based payload inspection — with <2 ms rule-evaluation latency.

CERTIFICATIONS

AWS Solutions Architecture Job Simulation

Forage · Jan 2026 · Cloud Architecture, Scalable Systems, Hosting Infrastructure

JPMorgan Chase Software Engineering Job Simulation

Forage · Jan 2026 · Kafka, REST APIs, Backend Integration

Deloitte Data Analytics Job Simulation

Forage · Jan 2026 · Data Analytics, Forensic Technology, Business Insights

TATA Data Visualisation: Empowering Business with Effective Insights

Forage · Jan 2026 · Data Visualization, Business Analysis

EDUCATION

Bachelor of Engineering — Computer Science and Engineering

CGPA: 7.8 · 2022 – 2026

College of Engineering and Technology, Akola, Maharashtra

- **Capstone project:** CDSS — AI-powered Clinical Decision Support System designed to improve patient analysis and healthcare.
- Engineered a 3-model ML pipeline (Logistic Regression, Decision Tree, Random Forest) to predict Covid-19 patient outcomes from symptom data; Random Forest achieved 82% accuracy, outperforming baseline Logistic Regression by 3 percentage points.